

The logo consists of a solid blue square. Inside the square, the letters 'EC' are positioned above the numbers '012'. Both 'EC' and '012' are rendered in a light blue, sans-serif font. The 'E' and '0' are slightly larger than the 'C' and '12' respectively.

EC
012

Consumer and producer surplus

Chapter 4

Where are we?

- We just covered **twelve principles of economics (CH1) semi - asynchronously**
- We learned how to set up **supply and demand curves** and **move them** around **(CH3)**.
 - We did the **double shifts** asynchronously

What questions do you have?

Help me help you

- Throw in any general topics where you struggled
- Any specific HW/LC questions you got stuck on?
- Throw in any admin questions
- Like/dislike anything that resonates with you

How is the pace so far?

a) Too slow

b) Just right

c) Too fast

d) 

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EC 012

Plan

Today

- A bit of review/recap
 - Focus on opportunity cost
- Work with some PS and CS

Thursday

- More review of demand shifts
- PS/CS

Opportunity cost

You've been dating Toby for two years. Toby is an extremely reasonable human and enjoys tacos, sledding, and sweet potatoes. Toby does not make your heart sing.

What is the opportunity cost of dating Toby?

Opportunity cost, take 2

You bought a Groupon (HA, remember those?) to a spinning class. It expires today!

There is only one more spin class offered tonight at the gym, and but you had planned to watch Billie Eilish host SNL



What best represents the opportunity cost of the spin class?

- a) Value of the spin class
- b) Value of the spin class + enjoyment of SNL
- c) No opportunity cost – you already paid for the class!
- d) Enjoyment of SNL

What would be the best way to regulate kidney donation in the US?

As is: voluntary w/ no compensation

Light modification: allow coverage of medical expenses

Iran model: Allow cash payments w/in gov. regulated market

Free market: Buy and sell

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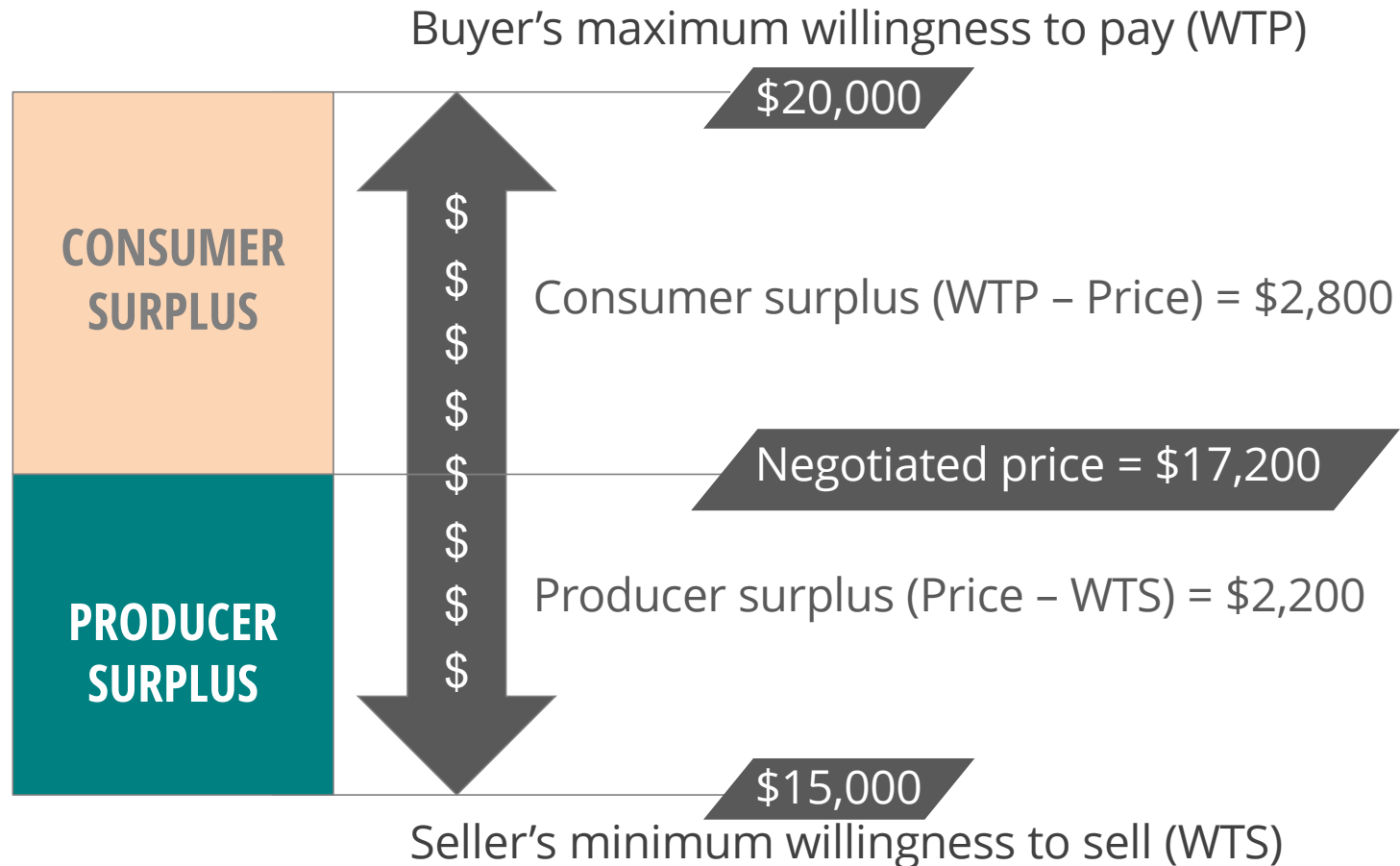
Free market: Buy and sell

Learning objectives

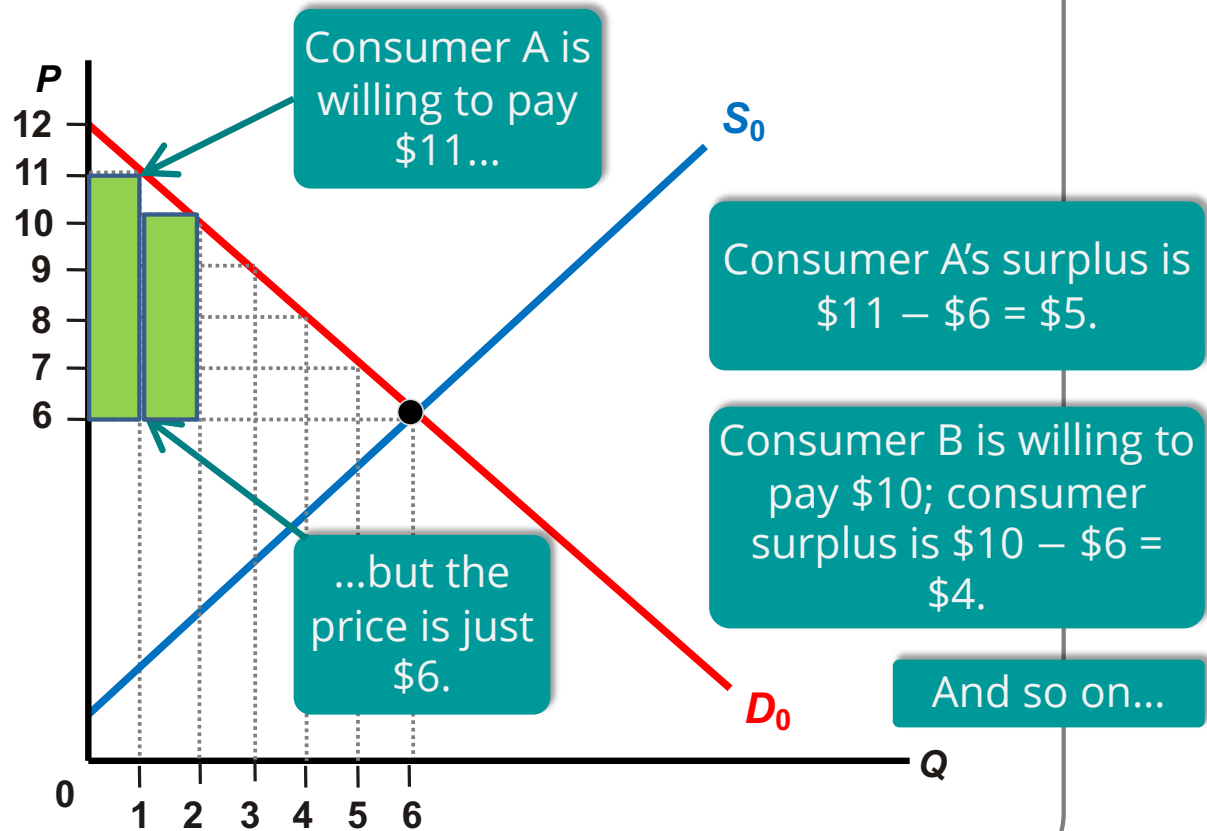
- Calculate **consumer** and **producer surplus**
- Define and identify **deadweight loss**
- When markets lead to **efficiency**
- Types of **market failures**

**CONSUMER
& PRODUCER**

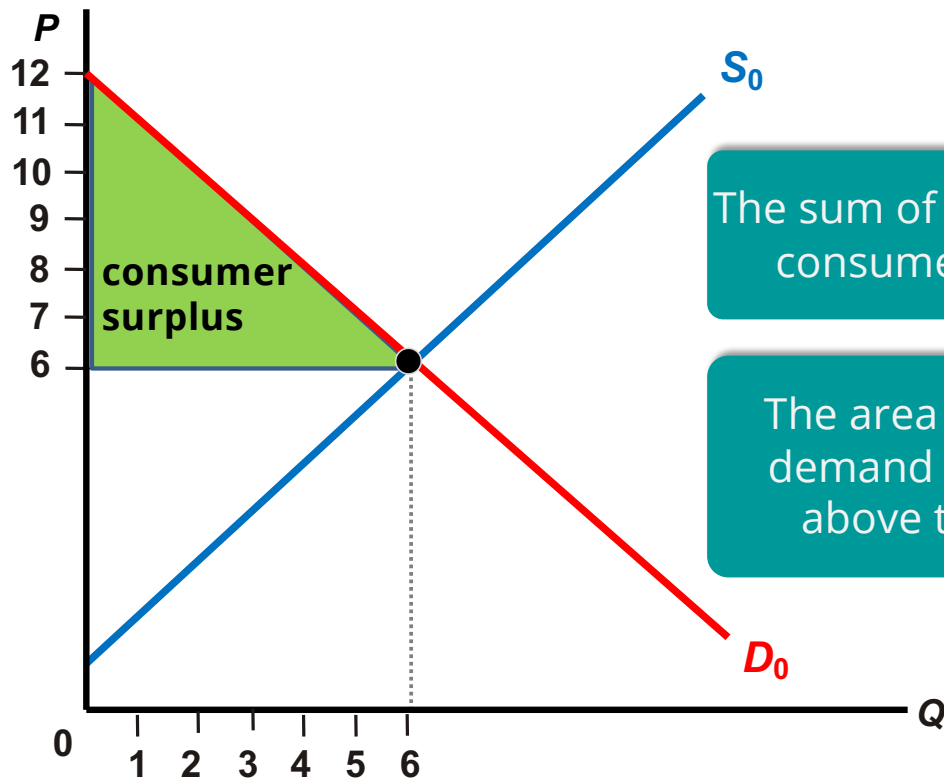
SURPLUS



INDIVIDUAL CONSUMER SURPLUS



TOTAL CONSUMER SURPLUS



The sum of all individual consumer surplus

The area below the demand curve and above the price

Buy some new shirts, George!

George wants to buy some new shirts for work. He is willing to pay \$35 for the first shirt, \$25 for the second shirt, and \$15 for the third.

Oxford & Co. (his favorite shirt shop!) sells shirts for \$28 each.

What is the efficient number of shirts for George to buy?

- a) 0
- b) 1
- c) 2
- d) 3



Buy some new shirts, George!

George wants to buy some new shirts for work. He is willing to pay \$35 for the first shirt, \$25 for the second shirt, and \$15 for the third.

Oxford & Co. (his favorite shirt shop!) sells shirts for \$28 each.

What is George's consumer surplus?

- a) \$0
- b) \$7
- c) \$10
- d) \$28
- e) \$35



Calculating consumer surplus

Mark and Rasheed are buying new calculators for Math 19. Mark is willing to pay \$75 and Rasheed is willing to pay \$100 for a graphing calculator. The price for a calculator at the bookstore is \$65. How much is their total consumer surplus?

- a) \$10
- b) \$35
- c) \$45
- d) \$60

Figure: The Market for Hamburgers



If the price of hamburgers is \$2, consumer surplus will equal:

- a) \$650
- b) \$400
- c) \$225
- d) \$450

If the price of hamburgers is \$2.50 today, what will happen to the price in the coming days/months?

- a) It will fall
- b) It will rise
- c) It will remain the same
- d) Not enough information

If the price of oil rises, what will be the main effect on the chart?

Figure: The Market for Hamburgers



- a) Supply will shift left
- b) Supply will shift right
- c) Demand will shift left
- d) Demand will shift right

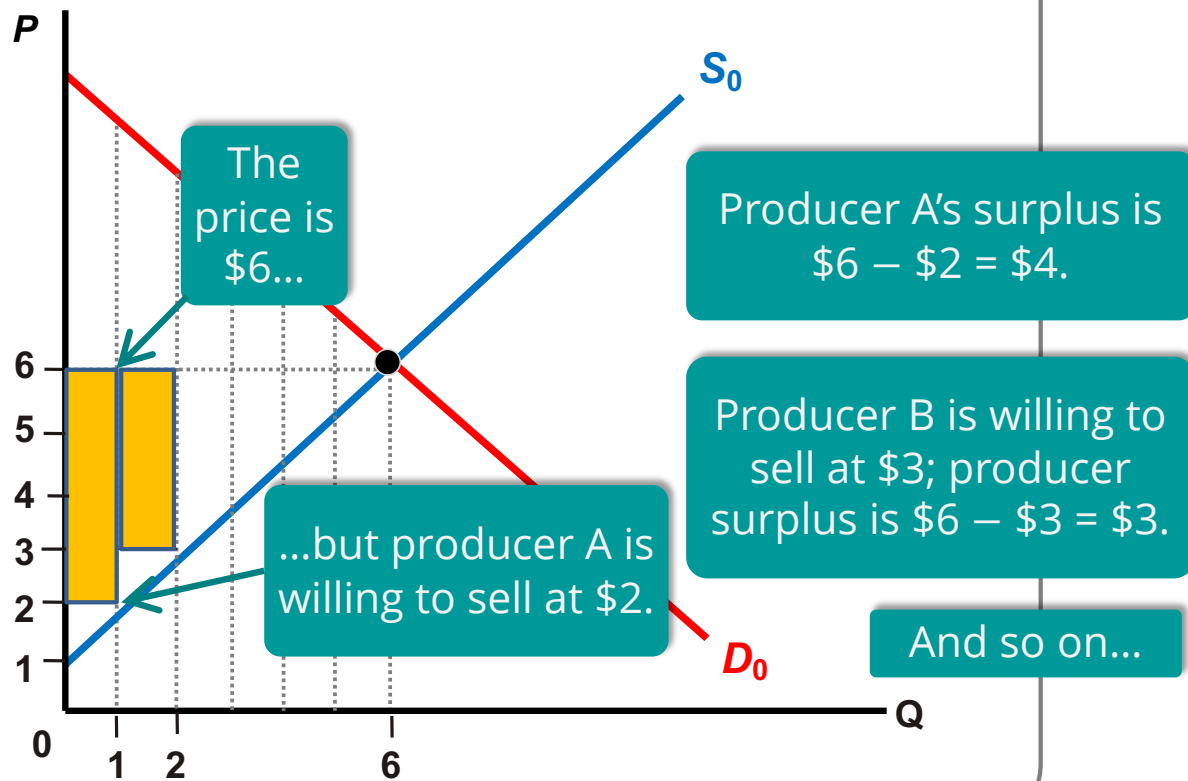
If the price of a hamburger is \$2, consumer surplus will be ...

Figure: The Market for Hamburgers



- a) \$650
- b) \$450
- c) \$400
- d) \$225

INDIVIDUAL PRODUCER SURPLUS



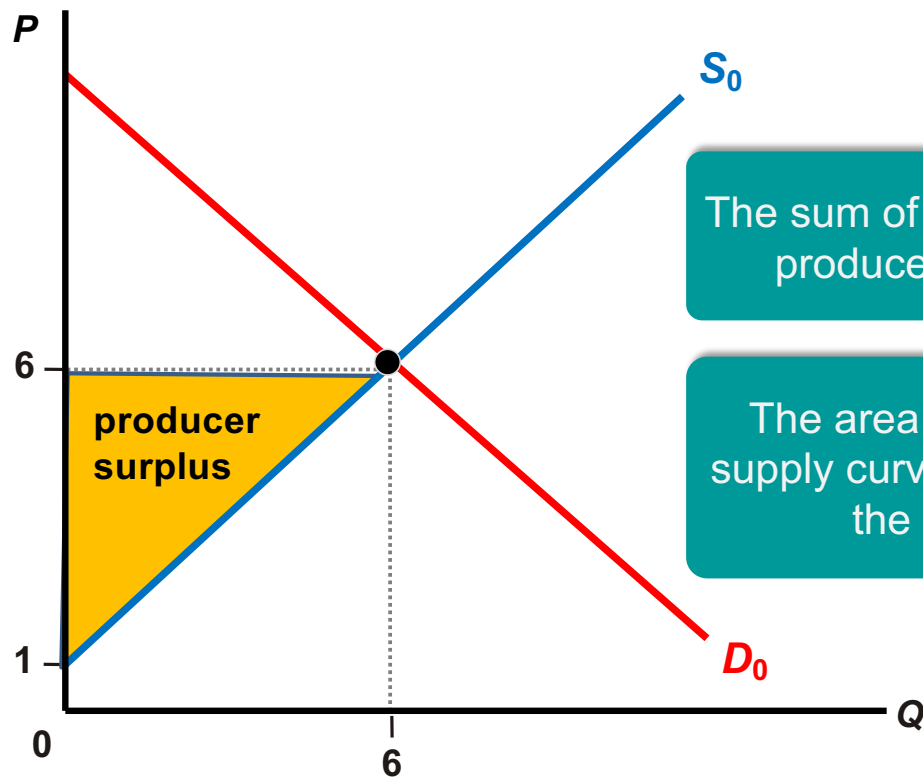
Four different students each have one textbook they might sell back to the bookstore. Below is each student's reservation price: the lowest price at which they would be willing to sell the book. The bookstore is buying textbooks at a price of \$25.

Seller	Seller's Reservation Price
Alex	\$12
Piper	\$16
Daya	\$22
Red	\$30

How many students will sell their textbooks?

- a) 0
- b) 1
- c) 2
- d) 3
- e) 4

TOTAL PRODUCER SURPLUS



The sum of all individual producer surplus

The area above the supply curve and below the price

Figure: The Market for Hamburgers



If 400 hamburgers are sold, producer surplus will equal

a) \$650

b) \$400

c) \$510

d) \$200

Figure: The Market for Hamburgers



If the price falls from \$1.50 to \$1, what is the loss in producer surplus?

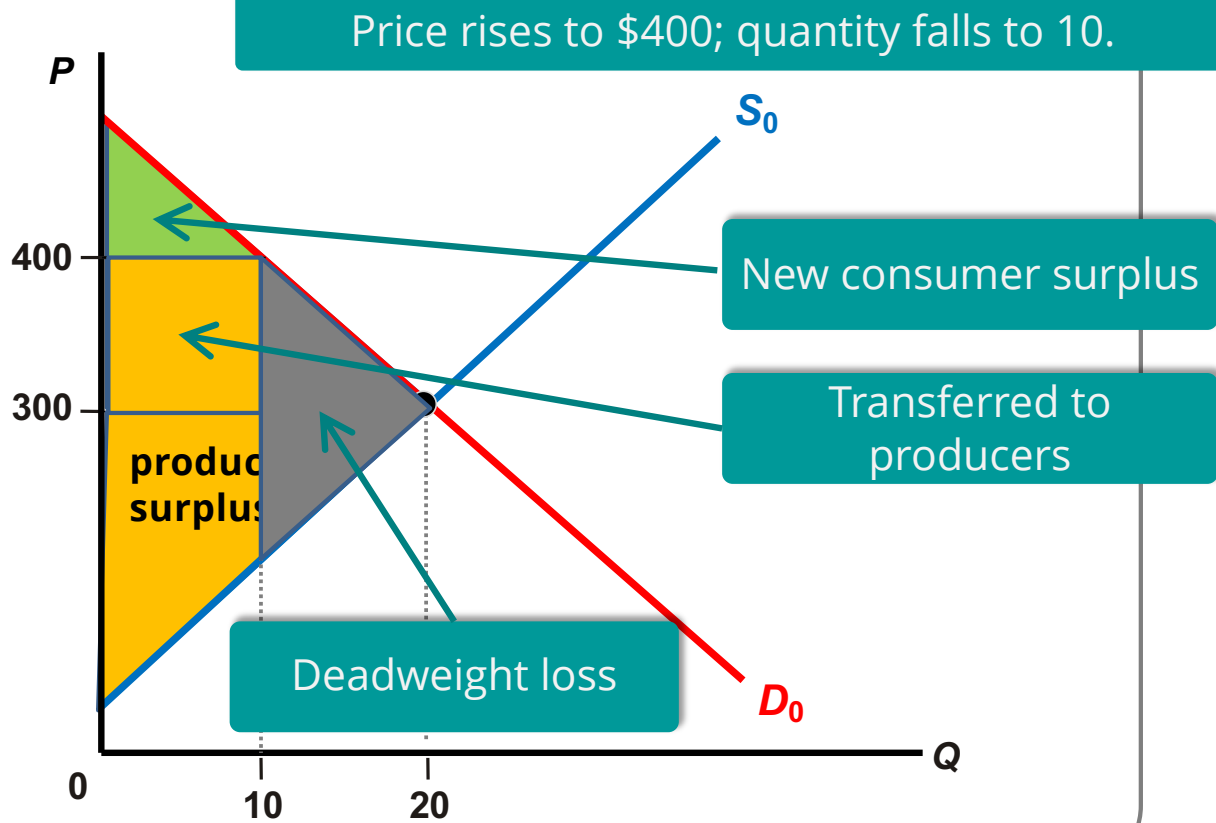
- a) \$50
- b) \$45
- c) \$150
- d) \$200

TOTAL SURPLUS

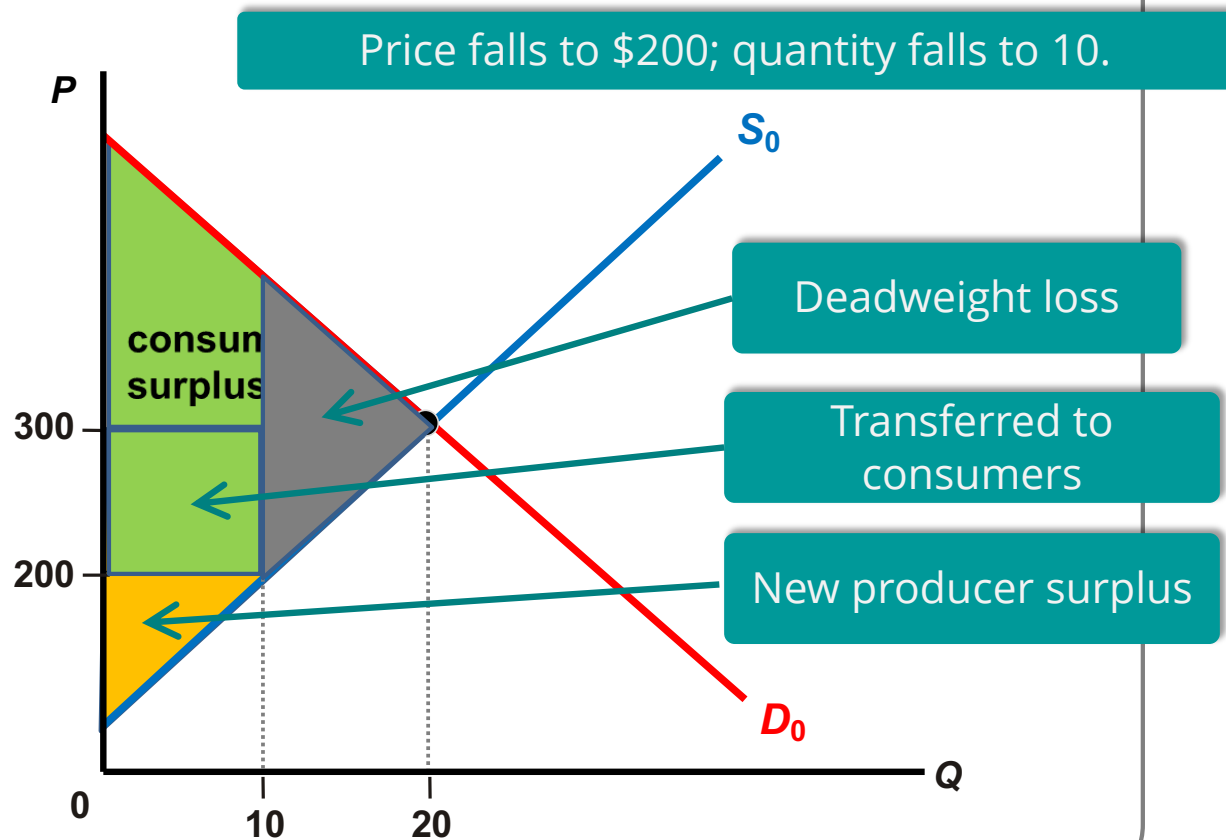
When markets are efficient,
TS is MAXIMIZED!

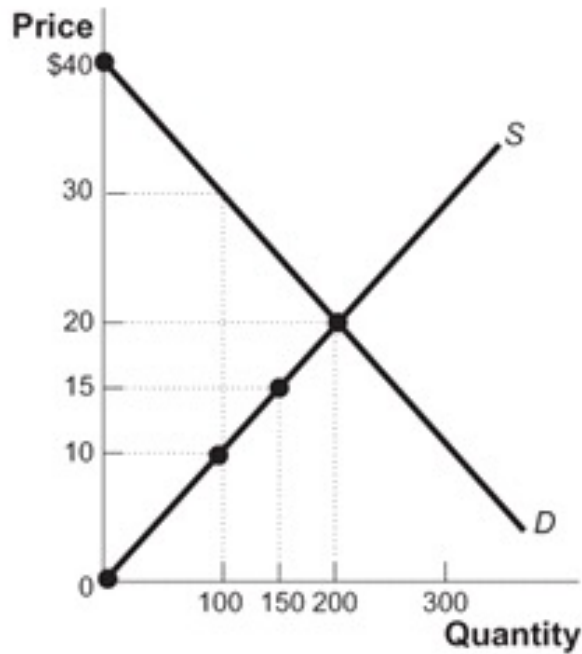
$$TS = CS + PS$$

WHEN PRICES EXCEED EQUILIBRIUM



WHEN PRICES ARE BELOW EQUILIBRIUM

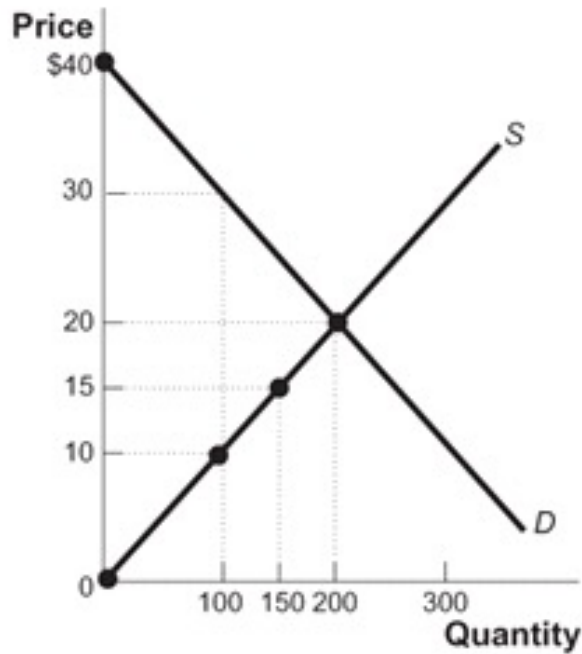




Look at the market for octopus candle holders. What is **producer surplus** at equilibrium?

- a) \$1000
- b) \$2000
- c) \$4000
- d) \$8000

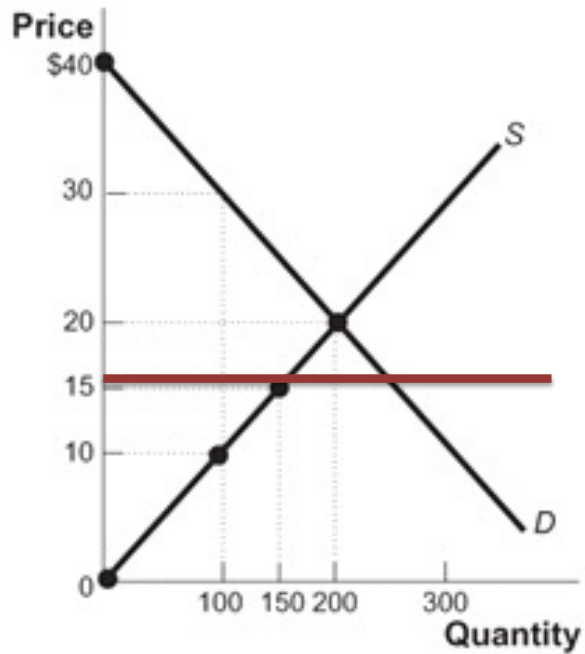




Look at the market for octopus candle holders. What is **total surplus** at equilibrium?

- a) \$1000
- b) \$2000
- c) \$4000
- d) \$8000





Amazon caps the price at \$15. What is the new **producer surplus**?

- a) \$1125
- b) \$1875
- c) \$2000
- d) \$2250



Efficiency of Markets

- Competitive markets are usually efficient:
 1. Allocate consumption to the potential **buyers who most value it.**
 2. Allocate sales to potential **sellers** who most value right to sell the good (e.g., **who have the lowest cost**).
 3. They ensure that **all transactions are mutually beneficial**

Why markets *typically* work so well

Well-functioning markets are effective because of:

1. **Property rights:** the rights of owners of valuable items, whether resources goods, to use those items as they chooseor.
2. **Economic signals:** any piece of information that helps people make better economic decisions.

Four sources of market failures

Markets will not be efficient in the presence of **market failures!**

- Lack of competition
- Information not shared by all parties
- Existence of external benefits/costs (externalities)
- Existence of public goods